

EXAM 7 – SPRING 2015

12. (1 point)

Given the following ground-up information for insurance claims subject to a single excess-of-loss reinsurance treaty as of December 31, 2014:

<u>Claim Number</u>	<u>Occurrence Number</u>	<u>Date of Loss</u>	<u>Reported ALAE</u>	<u>Reported Loss</u>	<u>Total Reported Loss and ALAE</u>
1	1	Feb 29, 2013	\$12,000	\$80,000	\$92,000
2	2	May 28, 2013	18,000	145,000	163,000
3	3	Sep 5, 2013	8,000	70,000	78,000
4	3	Sep 5, 2013	3,250	30,000	33,250
5	4	Jan 24, 2014	9,000	90,000	99,000
6	5	Mar 7, 2014	20,000	225,000	245,000
7	6	Oct 18, 2014	3,750	55,000	58,750

<u>Accident Year</u>	<u>Historical Per-Occurrence Retention</u>
Jan 1, 2013 – Dec 31, 2013	\$150,000
Jan 1, 2014 – Dec 31, 2014	200,000

- The historical pre-occurrence retentions apply to combined loss and ALAE.
- The reinsurance treaty stipulates that the reinsurer must be notified of any occurrences whose reported loss and ALAE reaches 50% of the retention.

Demonstrate how the data above indicates the presence of two aspects of reinsurance loss reserving that make it somewhat more difficult than primary loss reserving.

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QUESTION 12																																	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: A11: Compare and contrast reinsurance and primary reserving procedures.																																
SAMPLE ANSWERS																																	
Candidates needed to demonstrate the presence of two of the following three issues:																																	
Issue 1: Claim report lags to reinsurers are generally longer. <i>Sample Answer 1</i> Reinsurer would not know about claims 5 and 7. Claim 5 is likely to breach 50% of retention threshold and reinsurer would probably know about it a year later. <i>Sample Answer 2</i> Slow reporting lag – Reinsurer gets notified when Loss & ALAE hits 50% of retention which may take a while to develop (i.e. talc mass tort claims). They take a long time to develop (especially losses) so the primary might know of it way sooner than reinsurer.																																	
Issue 2: There is persistent upward development of most claim reserves (often due to tendency to underestimate ALAE). <i>Sample Answer 1</i> Ratio of ALAE to Loss <table> <tr> <th><u>Claim</u></th><th><u>Ratio</u></th></tr> <tr><td>1</td><td>0.15</td></tr> <tr><td>2</td><td>0.1241</td></tr> <tr><td>3</td><td>0.1143</td></tr> <tr><td>4</td><td>0.1083</td></tr> <tr><td>5</td><td>0.1</td></tr> <tr><td>6</td><td>0.0889</td></tr> <tr><td>7</td><td>0.0682</td></tr> </table> There's an upward development of ALAE because ALAE tends to be under-reserved. <i>Sample Answer 2</i> ALAE / Loss <table> <tr> <th><u>Claim</u></th><th><u>Ratio</u></th></tr> <tr><td>1</td><td>0.15</td></tr> <tr><td>2</td><td>0.124</td></tr> <tr><td>3</td><td>0.114</td></tr> <tr><td>4</td><td>0.108</td></tr> <tr><td>5</td><td>0.1</td></tr> <tr><td>6</td><td>0.089</td></tr> <tr><td>7</td><td>0.068</td></tr> </table> Decreasing trend as you look down the table – There tends to be an upward development of reserves as primary insurers tend to underestimate LAE costs. AY 2014 claims have lower ALAE-to-Loss ratios than AY 2013, so they may be understated.		<u>Claim</u>	<u>Ratio</u>	1	0.15	2	0.1241	3	0.1143	4	0.1083	5	0.1	6	0.0889	7	0.0682	<u>Claim</u>	<u>Ratio</u>	1	0.15	2	0.124	3	0.114	4	0.108	5	0.1	6	0.089	7	0.068
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Issue 3: Heterogeneity of patterns. <i>Sample Answer 1</i> Each accident year has different retention, making patterns different and adding difficult																																	

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since there are only two accident years.

Sample Answer 2

Reinsurers have very heterogeneous exposures. Exposures vary by LOB, contract terms, etc. The retention changed in 2014, so historical losses before 2014 can't be directly used to compare with losses after 2014.

EXAMINERS' REPORT

- Candidates were expected to have an overview of reinsurance and primary insurance reserving methods, the assumptions underlying them, and how those assumptions could be violated due to differences in information available to reinsurers.
- Candidates often identified common issues with reinsurance data, but then had difficulty using the provided data to demonstrate the presence of these issues.
- The most common error was misinterpreting the information provided in the question as an actual report provided to the reinsurer. Therefore many candidates used "the reports the reinsurer receives may be lacking some important information" as an issue.
- Another common error was stating "there is a persistent upward development of most claim reserves" often due to the "tendency to under-reserve ALAE", but then not using the individual claims to show how the ALAE ratios were increasing with the age of the claim.